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CENTRE FOR INNOVATION, INCUBATION AND ENTREPRENEURSHIP

VENTURER'S VOYAGE

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INTRODUCTION

Agriculture sector is an important sector in India with about 45% of the population (According to World Bank ILO Estimates) of India being employed(directly) by the same but generating only 20.2 percent in GDP yield.

This is changing, as innovation is changing the sector of agriculture and associated sectors as more modern ways of farming, making new and more useful products of agriculture and techniques to grow more in limited land are coming into the market. The Indian government and entrepreneurs are also going and learning from foreign nations who are good at one or another aspect of farming and making products related to it and bringing in new products in the Indian market and improving the state of agriculture in India. Such products are or will definitely help to improve the current state of the industry and the ones related to it.

Following we have some articles where we show such innovative Ideas and even some Implementation that are being done in the industry.

Hope all of you find the articles informative and they ignite a spark in you for entrepreneurship especially in the agriculture sector.

Happy Reading.

PEARLS OF THE FARMERS

Since agriculture is the foundation of the Indian economy, a strong, research-backed seed distribution is necessary. The introduction of new goods has made the Indian seed market unstable at the moment, but it is expected to expand at a CAGR of 6.8% through 2025. The market is fiercely competitive, and businesses with a creative mindset and the capacity to deliver a product that meets farmers' expectations have a higher chance of surviving. In 1885, Namdeo Umaji Agritech was founded in a modest shop in Byculla, Mumbai, and is today well-known on the global market. It is also acknowledged as India's oldest continuously operating seed business.

"We are pioneers in the distribution of vegetable seeds and were one of the first firms to import hybrid vegetable seeds to India in 1991," says Shlok Bhalinge, Director of Namdeo Umaji. We launched our own seed company in 2001 under the name Pyramid Seeds. The company's attitude to changing in line with the demands of the Indian market, particularly its capacity to adapt biotechnology and contemporary plant breeding techniques as the newest trendsetters in agriculture, is what sets it apart.

Realizing Customer Expectations

Offerings from Namdeo Umaji include vegetable seeds, herbs, and lower seeds; these aren't always field crops like paddy or rice. We have a Research & Development centre that is registered with DSIR, Shlok continues. In order to change the industry and bring better crop types to the Indian market, we continue to do market research while also breeding our own kinds. The newest kinds provide farmers a better yield and disease resistance to secure the farmers' profit all season long.

In order to create better goods for the Indian market, we gather germplasm from all over the world and employ our R&D facilities, bearing in mind the products' capacity to respond to climate fluctuation, soil texture, and customer demand. According on the type of crop, this aids farmers in assuring the quality of the crops in terms of colour, shape, size, nutritional value, etc. Our business is committed to using its decades of expertise in the agriculture sector to study and produce the next product that the market demands.

PROVIDING FARMERS WITH LONG-TERM GAINS

A fantastic sales and marketing staff at Namdeo Umaji focuses on building relationships with farmers. With the help of local sales professionals, the organisation examines and addresses each of its customers' difficulties. "I completed my undergraduate degree at UC Davis, the top agricultural institution in the world, and am currently pursuing a master's degree there, adds Shlok. I completed many internships and accumulated 2-3 years of professional job experience. Ugaoo.com, a website with A-Z Gardening needs and all of our knowledge and seeds in its product profile, was founded by my older brother. He is in charge of organic farming, terrace farming, and the urban market in general.

MAP FOR THE FUTURE

Because newsletters are easily sent online, you can weave all of your important and latest services in one go. Include daily updates or a weekly round-up that's composed of concise information about the nature of your business. Whether you're in tech or the entertainment industry, craft interesting content that your customers will enjoy at first glance.

Shlok Bhalinge
Director of Namdeo Umaji
Agritech India pvt. Ltd.

AGRICULTURE AND INNOVATION

Agriculture is the backbone of the Indian economy. The country produces 11% of total global agricultural while also housing the world's biggest population of malnourished people.

Agriculture provides us with healthy, safe, and nutritious food, but present production risks depleting and harming the natural resources on which it is based. Climate change and new threats to global food systems must also be addressed by the sector.

Innovation enables us to accomplish more and get higher quality results with fewer resources. Many advances in agricultural technology are "process innovations" that enhance production methods, such as improved seed varieties or more effective irrigation systems. Items like healthier foods, as well as new chemical and pharmaceutical products, are examples of what are known as "product innovations,". Supply chains increasingly rely on "marketing and organisational innovations."

The digital transformation and innovation in agriculture currently occurring worldwide holds some promise for progress, but given the complexity of Indian agriculture, no single policy change or technological shift will be enough to achieve the country's dual goals of increasing income for smallholder farmers and continuing to strengthen the competitiveness of Indian agriculture.

Many areas of agriculture have benefited from digital transformation, including product marketing, information assessment, issue solving at multiple levels, improved communication, access to capital, and many more.

Effective improvements in Indian agriculture have given the industry a fresh hope for the future. Vertical farming, genetic editing, and the use of technology such as artificial intelligence and IoT to improve things have all greatly aided Indian agriculture. Scientists have also begun to develop crops that can grow on less water while producing more food.

Effective improvements in Indian agriculture have given the industry a fresh hope for the future. Vertical farming, genetic editing, and the use of technology such as artificial intelligence and IoT to improve things have all greatly aided Indian agriculture. Scientists have also begun to develop crops that can grow on less water while producing more food. Waste management can also be approached differently in agriculture to provide economic benefits. Many small businesses make inventive items from agricultural waste, such as colours, sugar, essence sticks, antibiotics, biofuels, MDF boards, and packing materials.

This is a great step in implementing circularity in agriculture and generating new, more effective methods for managing our limited natural resources.
The youth of India hold the key to bringing about a multifaceted agricultural revolution through their innovations at all levels.

Dr. Rohini Sawalkar - Saxena
Faculty, MIT-WPU

AGRICULTURAL ROLE IN INDIAN DEVELOPMENT

Agriculture is a very common word which is used almost by everyone. Agriculture plays a very important role in living life. It is impossible for us to sustain life without agriculture.

Agriculture means Sow Art and Science in soil growing crops and raising livestock. When the British were ruling over their Indians the British were forcing the farmers to grow the same crops as the British told them after India became independent in 1947 the farmers got the freedom to farm in any way they wanted.

In Modern era, there are so many innovations introduced in market, but because of global warming farmers Sold His agriculture land and shift in 2 cities decreasing a rainfall is the main effect on agriculture sector and due to the lack of water the farmers cannot cultivate and the farmers who are doing agriculture are harvesting rainwater and doing agriculture and this is the basic innovation in agriculture sector.

Due to this reason India was once known as an agricultural country, but in the present Times India is known for service sector based because nowadays service sector is contributing in GDP is 53.8 89% and Agriculture sector is contributing is only 14%.

Problem & Solution:

- At present there is a lot of good technology for farmers in the market. For example, Good machines have come, drones have come but farmers are not aware of the technology. We can educate the farmers about the technology.
- In the present time Indian government has launched a menace good scheme for farmers, but farmers are not able to take advantage of this due to lack of knowledge of the scheme. We can make farmers aware about that scheme.
- At present many innovations related to agriculture have come to the market. But this innovation has not reached the farmers. We can educate the farmers of those innovations and make them aware of how those innovations are beneficial to them.
- Many farmers are lacking from a financial aspect, low interest rates are being used by well to do families and the benefits are getting reached with the farmers. We can design a way which the app would focus the real farmers who are needy and provide them funds at very less rate.

Omkar A. Patil (FY MBA Div. D)
Marketing + Sports Management

CROP SENSOR AND AGRICULTURAL ROBOTICS

Innovation in agriculture has a long and storied history. From the development of irrigation systems and crop rotation in ancient times to the modern-day advances in biotechnology and precision farming, farmers have always been at the forefront of developing new ways to increase yields and improve the quality of their crops. Today, the challenge for farmers is to keep up with the ever-changing landscape of agriculture. With the global population projected to reach 9.7 billion by 2050, the food and agriculture sector will need to increase production by 70 percent to meet the demand. This will require major advances in technology and management practices across the entire food system. precision farming, a set of management practices that use information and technology to optimize crop production, is one promising area of innovation. In addition to precision farming, there are a number of other innovative technologies and practices that are being developed to help meet the challenges of the future. One of them is crop sensors. Crop sensors help in fertilizer application efficiency and absorption. They can monitor how your crop is feeling and help to reduce the danger of leaching and discharge into groundwater. Rather than developing a prescription fertilizer plan for a field before heading out to apply it, crop sensors advise application equipment what amount to distribute in real-time. Optical sensors can detect how much fertilizer a plant needs based on the amount of light reflected back to the sensor.

Second one is Robots in Agriculture this is the sector where innovation is going to take place in future. This will be going to a big market in India for the next couple of years. agricultural robots automate slow, repetitive and dull tasks for farmers, allowing them to focus more on improving overall production yields. There are some applications of agriculture robotics and some of the applications are Crop and Soil Monitoring, Predictive Agricultural Analytics, Supply Chain Efficiencies, Agricultural Robots and etc. these are some applications of robotics in agriculture.

Prathamesh Kulkarni (FY MBA Div. 12)
Marketing

AQUATILAGE- REDEFINING FARMING

Aquatilage works on the principle of Aquaponics. It is the integration of recirculating aquaculture and hydroponics in one production system. Aquaponics is resilient because of its capacity to adapt to local ecological constraints. According to historical data, aquaculture and hydroponics have been used in tandem since 2000 B.C. Aquaponics was used by various cultures, including the Japanese, Peruvians, and Greeks, even though the Aztecs, Egyptians, and Chinese are known for using it most famously in history. Recent studies also contend that aquaponics was used to maintain Babylon's hanging gardens. Aquaponics has historically offered metropolitan centers facing severe population constraints agricultural solutions.

Recirculating aquaculture and hydroponics are combined in an aquaponics system to produce food. At Aquatilage with the help of aquaponics farms we create very environmentally friendly farming since it makes the most of all resources, particularly water. An aquaponic farm uses water that is continuously recycled from a tank using biofilters, and the nutrients are fed to the produce.

Aquaponics uses 90% less water than conventional soil farming and can produce more nutrient-dense organic food than any other method ever could.Spaces as big as abandoned buildings or as tiny as the alley behind the house can be used to set up aquaponics farms. This might be the answer to the management of water and the new agricultural revolution.

Here is how we do it: water from the fish tank circulates through filters, plant grow beds, and back to the fish in an aquaponic system. The fish wastes are removed from the water in the filters by first going through a mechanical filter to get rid of the solid waste and then a biofilter to get rid of the dissolved wastes. Ammonia, which is poisonous to fish, is changed by bacteria in the biofilter into nitrate, a more readily available fertiliser for plants. Nitrate and other nutrients are absorbed by plants when the water flows through plant grow beds, purifying it before it is returned to the fish tank.

VIVEK SINGH (SY-MBA, Div -A)
MBA INTERNATIONAL BUSINESS

MULCH FILMS

Plastic mulch may be a product used in plasticulture to suppress weeds and conserve water. Certain plastic mulches also act as a barrier to stay methyl bromide, both a strong fumigant and ozone depleter, within the soil. Many acres are cultivated under plastic mulch worldwide each year.

Mulch films are wont to modify soil temperature, limit weed growth, prevent moisture loss, and improve crop yield also as precocity. Due to their thickness, use of pigments and their exposure to high solar irradiation, mulch films require proper light and thermal stabilizers with intermediate chemical resistance.

Mainly used for growing food and vegetable crops over extensive areas, much films offer the subsequent

Advantages:

* Improved crop quality
* Water retention
* Minimization of weed spread
* Soil temperature control
* Soil disinfection before sowing
* Reduction in root damage

Disadvantages:

Cost
Environmental concern
Application
Planting

Disposal of plastic mulch is an environmental problem. Technologies exist to supply for the recycling of used/disposed plastic mulch into viable plastic resins for re-use in the plastics manufacturing industry. However these methods aren't very effective due to contamination by agrochemicals of the plastic. Other concerns include residual microplastics within the soil which can have negative effects on soil ecologies, including microbes and earthworms.

Other concerns include residual microplastics within the soil which can have negative effects on soil ecologies, including microbes and earthworms.
Main sorts of mulching
* Clear mulch (transparent)
* Black mulch (total opacity)
* Silver/Black or White/Black mulch (bi-colour)
* Metallic mulch
* Photo selective mulches (Brown or Green)
* Biodegradable mulching

TYPES OF MATERIALS USED FOR PLASTIC AGRI FILM:-
* LDPE/LLDPE: Mono-layer or co-extruded polyethylene. This is often considered the most standard type of film found in the US.

* HDPE: High density blended films which are stronger than traditional LDPE. they supply greater strength at lower gauges but they also are stiffer and more difficult to lay.
* Metalized: Usually LDPE films that are coated with a microscopically thin layer of pure aluminium. The quantity of aluminium used is so small that it is less than 1% of the weight of the roll.
* VIF: This is usually a 5 layer film where the outer two layers are LDPE. The innermost layer is nylon and it's surrounded by a tie layer on each side.
.
TIF: TIF's structure is usually the same as VIF except the center layer which produces the barrier is made of EVOH instead of nylon

ANIKET SARDA (FY-MBA DIV-D)
MARKETING

AGRICULTURAL INNOVATION IN INDIA

SNEAK PEEK OF WHAT'S INSIDE:

FUTURE IS BRIGHT FOR INDIA IN DRONE TECHNOLOGY

According to reports, the agriculture industry in modern India contributes the most to GDP. 18% of India's GDP is directly attributable to its agriculture industry. One of the biggest producers of agricultural food on the global market is our country. Additionally, agriculture is regarded as the major source of income for around 58% of the people of the nation, primarily in rural regions. However, the agriculture industry is the most promising one. It has several challenges today, one of which is the lack of labour for farming, thus India is improving productivity and efficiency in the drone industry to realize its full potential. There are a number of aspects and issues that need to be recognized, supported, and addressed with solutions. A lot of critical farming practices, like the use of pesticides to manage pests and illnesses and monitor crops, are now done in an improper manner.

One of today's several issues is the scarcity of farm labour.

Why the future is bright for India in Drone Technology -:

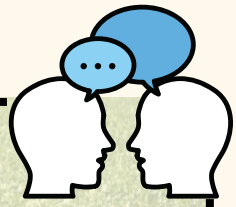
Drones have been more prevalent in people's lives during the last year. Drone technology has advanced significantly in India during the previous two years. Furthermore, many large-scale farms have begun to use drone technology. Because it provides tremendously beneficial benefits to farmers. Agricultural drones are gaining popularity because they can do a number of activities such as dusting pesticides, recording agricultural images using GPS technology, and providing information on water and fertilizer levels. Drones have the potential to improve agricultural operations in the areas of crop health assessment, irrigation, soil analysis, and field analysis.

- Furthermore, agricultural drones provide farmers with access to high-quality data that they may utilize to make better management decisions, boost crop yields, and increase overall profitability.
- Drones can be used to collect data on crop yields, animal health, soil quality, nutrient assessments, weather and rainfall patterns, and other topics
-

Rohit Chavan (FY MBA- Div 12)
International business

RIDDLES

1. I am referred to as Farmer's best friend. Who am I?
2. Who is the father of the Green Revolution in India?
3. Who gave the slogan "Jay Jawan, Jay Kisan"?
4. What fruit is green outside and green inside and has a big seed?
5. I am a seed with a 3 letter name. Take away the last two. I would still say the same.
6.Sweet water standing up. What is it?
7.Am referred to as white gold to farmers. Who am I?



JARGONS

Agronomic data

Data derived from activities and conditions on farm fields. The data can include soil analysis, nutrient information, hybrid selection, plant populations, and yield data.

Aquaponic

The integration of aquaculture (growing fish for food) and hydroponic vegetable production; these systems rely on plants to absorb fish waste from the water, which substitutes as fertilizer

Commodity Crop

This term generally refers to crops that are traded on international markets; corn, soybeans, wheat, rice and other grains are among the most common commodity crops

AgriStack

The Ministry of Agriculture and Farmers Welfare has planned creating 'AgriStack' - a collection of technology-based interventions in agriculture.

Unified Farmer Service Platform (UFSP)

UFSP is a combination of Core Infrastructure, Data, Applications and Tools that enable seamless interoperability of various public and private IT systems in the agriculture ecosystem across the country.

National e-Governance Plan in Agriculture (NeGP-A)

A Centrally Sponsored Scheme, it was initially launched in 2010-11 in 7 pilot States, which aims to achieve rapid development in India through use of ICT for timely access to agriculture related information to the farmers.

GAP

An acronym for Good Agricultural Practices, a voluntary protocol aimed at improving food safety; the USDA and other agriculture agencies offer GAP audits to verify that farmers and food processors are following the practices

1.Earthworm 2.S Swaminathan 3.Lal Bahadur Shashtri 4.Avocado 5.Pea
6.Sugarcane 7.Cotton

MEET THE COMMITTEE

Names	Designations
Evangeline Thomas	President
Baljeet Ahir	Vice President
Arpit Nema	General Secretary
Pragati Agrawal	General Advisor & Treasurer
Onkar Jadhav	Creative Head
Hrushikesh Shinde	Creative Co-Head
Shubham Maske	Sponsorship Head
Mrunal Mohan	Event Head
Saurabh Hole	Event Co-Head
Simran Shanker	Social Media Head
Disha Kaur Sawhney	Social Media Co-Head
Prasad Jagtap	Editorial Head
Disha Chhajed	Editorial Co-Head
Himanshu Tiwari	Technical Head
Shriniwas Nipanikar	Technical Co-Head
Aditya Jadhav	Promotions Head
Keely Carvalho	Promotions Co-Head

Under the Guidance of

PROF. DR. VINITA KALE — Coordinator, Center for Entrepreneurship Development

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Note: Articles should be on topics related to Entrepreneurship along with your name, specialization, year, contact number , email id and a formal photograph.
Word limit: 250-300 words.